# Golf Ball Speed Physics Research

Reference Document for Golf Video Game Development

Compiled from physics analysis, launch monitor data (TrackMan / Foresight), and simplified aerodynamic modeling — August 2026

## 1. Core Principle

Across **every club in the bag** (including the putter), the golf ball reaches its maximum speed at the instant it separates from the clubface. There is no post-impact acceleration that increases ball speed beyond the launch value. After separation, external forces (aerodynamic drag for full shots, or green friction for putts) cause the ball to decelerate.

**Practical rule for game physics:** Treat measured or calculated ball speed as the peak initial velocity vector. All subsequent forces only reduce speed (or, in rare descent cases for full shots, produce a minor recovery that is still far below launch speed).

## 2. Full Shots (Driver through Wedges)

Once the ball leaves the club, three primary forces act on it:

- **Gravity** — constant downward acceleration (≈ 9.81 $m/s^2$).
- **Aerodynamic drag** — opposes velocity; magnitude scales roughly with $speed^2$. Dimples reduce the drag coefficient in the relevant speed range, allowing the ball to retain speed longer than a smooth sphere.
- **Magnus (lift) force** — generated by backspin; acts roughly perpendicular to the velocity vector and keeps the ball aloft longer.

Because drag is strongest at high speed, the largest absolute speed loss occurs early in the flight. Near the apex, speed typically reaches a minimum. On the descent a modest recovery often occurs as gravity adds to the velocity magnitude while drag is lower, but landing speed remains substantially below launch speed (commonly 40–50% of launch for a driver).

### 2.1 Driver Example

A typical solid driver shot launches near 160–170 mph (PGA Tour average ≈ 167 mph). Speed drops rapidly in the first 1–2 seconds, reaches a minimum near or just after the apex, and lands in the 60–75 mph range in simplified models.

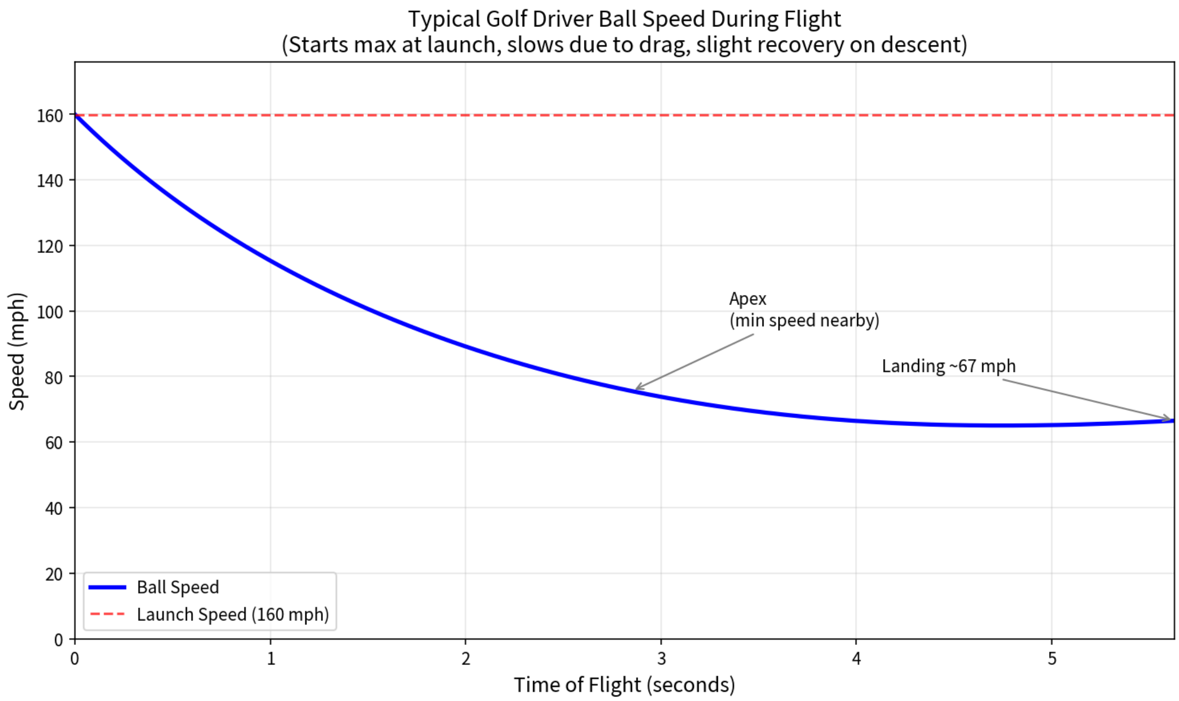


Figure 1. Typical driver ball speed vs. time of flight (simplified aerodynamic model).

## 3. Comparison Across Clubs

The same qualitative behavior holds from driver to pitching wedge: peak speed at launch, then continuous deceleration dominated by drag. Absolute launch speeds and trajectory parameters change with loft.

| Club | Typical Ball Speed | Launch Angle | Spin (approx) | Notes |
|---|---|---|---|---|
| Driver | ~167 mph | ~11° | ~2,700 rpm | Longest hang time, largest absolute speed loss |
| 3-Wood | ~158 mph | ~9–10° | ~3,600 rpm | Similar profile to driver |
| 7-Iron | ~120 mph | ~16° | ~7,000 rpm | Higher lift, steeper landing |
| Pitching Wedge | ~102 mph | ~24° | ~9,300 rpm | High spin, short air time |

Table 1. Representative PGA Tour-average launch conditions (TrackMan-sourced). Amateur numbers scale lower but follow the same relative pattern.

Higher-lofted clubs start slower, so the absolute speed drop is smaller, yet the percentage loss remains large. High spin increases early lift, which lengthens hang time relative to the lower launch speed, but drag still dominates the speed history.

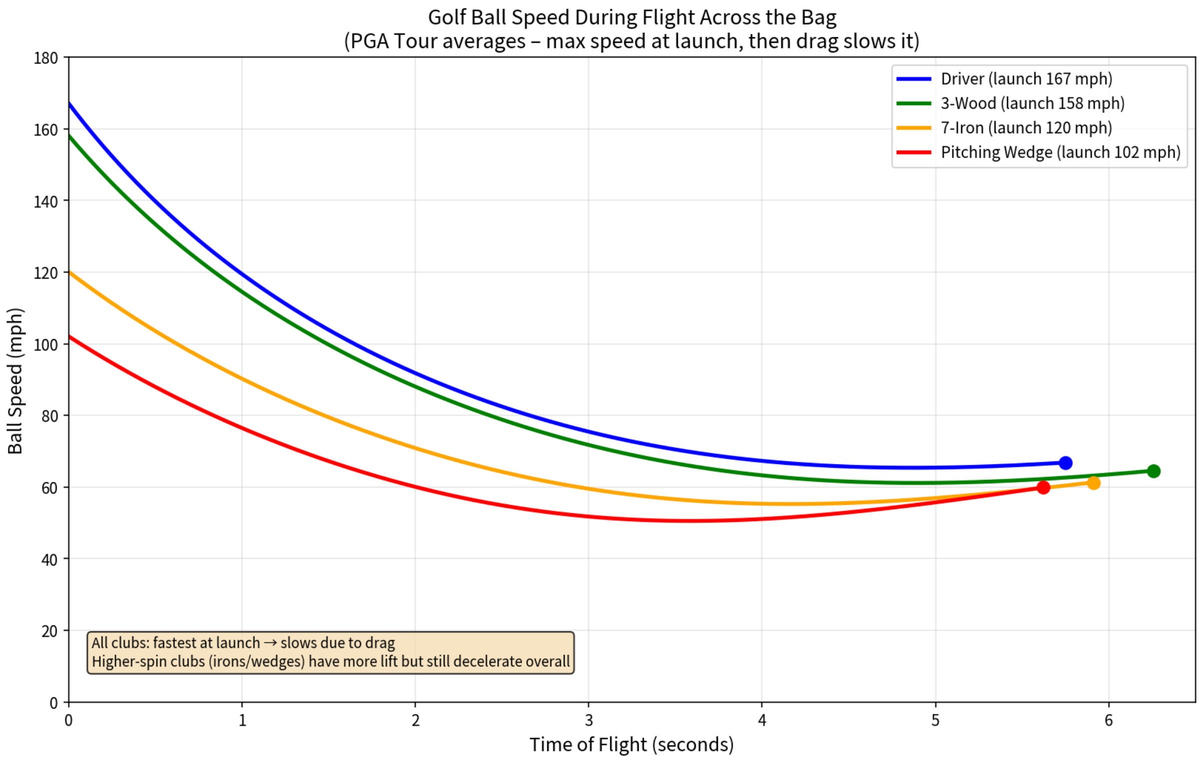


Figure 2. Ball speed vs. time for four representative clubs (same simplified model).

## 4. Putter — Ground Contact Physics

Putting is fundamentally different. The ball has essentially **no airborne phase**. After impact it skids briefly, transitions to pure rolling, and decelerates due to rolling friction with the green surface until it stops or enters the hole.

Key points:

- Peak speed still occurs at separation from the putter face.
- Typical initial speeds: 3–6 mph (short putts), 6–9 mph (medium), 9–12+ mph (long lags).
- Aerodynamic forces are negligible; the dominant force is green friction (rolling resistance).
- Deceleration is approximately constant on a flat green; slope adds or subtracts a gravitational component.
- Green speed (Stimp) directly controls the friction coefficient and therefore how quickly the ball slows.

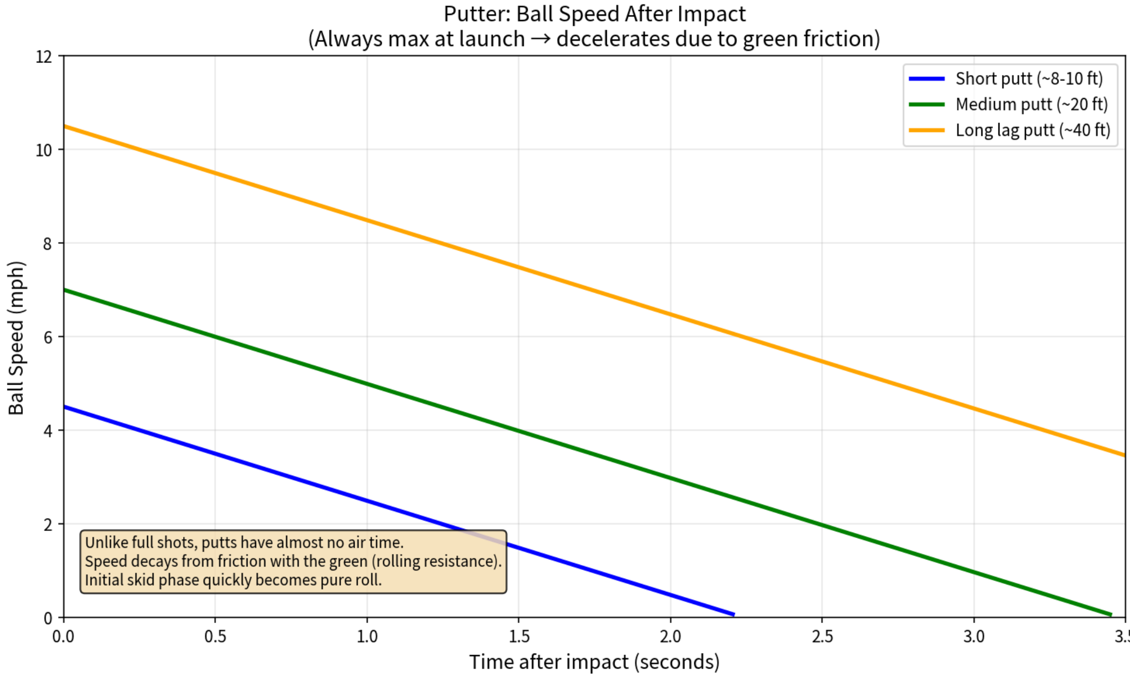


Figure 3. Approximate ball-speed decay for three putt lengths on a flat green (linear friction model).

# 5. Implementation Recommendations for Game Physics

## 5.1 Full-Swing Ball Flight

- Store the post-impact ball speed, launch angle, and spin as the initial state.
- Integrate forces each frame: gravity + drag (½ ρ Cd A $v^2$ opposite velocity) + Magnus lift (scaled by spin rate and approximately perpendicular to velocity).
- Cd is reduced by dimples (typical effective values 0.20–0.30 in the relevant Reynolds range). Cl scales with spin rate and decreases as spin decays (~3–4 % per second).
- Expect landing speed to be substantially lower than launch; use this lower energy plus remaining spin and landing angle to drive bounce/roll models.

## 5.2 Putter / On-Green

- Apply an initial linear velocity (and optionally a small vertical component for firm strikes).
- Model a short skidding phase that quickly converts to pure rolling (friction torque increases angular velocity until $v = \omega r$).
- Thereafter apply constant-magnitude rolling friction opposing velocity, modulated by Stimp and slope.
- Ball speed decreases monotonically; stop the simulation when speed falls below a small threshold or the ball is captured by the hole.

## 5.3 Club Scaling

Scale initial conditions by club: lower ball speed, higher launch angle, and substantially higher spin as loft increases. Smash factor also declines (driver ≈ 1.48–1.50, short irons/wedges lower).

# 6. Quick Reference Summary

| Aspect | Full Shots (Driver–Wedge) | Putter |
|---|---|---|
| Peak speed location | Instant of separation from clubface | Instant of separation from putter face |
| Primary slowing force | Aerodynamic drag (+ gravity) | Green rolling friction |
| Air time | Several seconds | Negligible / none |
| Typical peak speeds | 100–170+ mph | 3–12 mph |
| Landing / final speed | Much lower than launch (often 40–60 %) | Approaches zero (or hole capture) |

## Notes & Sources

Launch-condition numbers are drawn primarily from publicly reported TrackMan PGA Tour averages and Foresight Sports reference tables. Speed-vs-time curves were generated with a simplified constant-Cd / spin-scaled-Cl numerical integration for illustration; real aerodynamic coefficients vary with Reynolds number and spin rate. Putter friction models are consistent with published rolling-friction analyses related to Stimpmeter readings. This document is intended as a practical physics reference for game systems rather than a full aerodynamic treatise.